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MIL-PRF-22710/23C

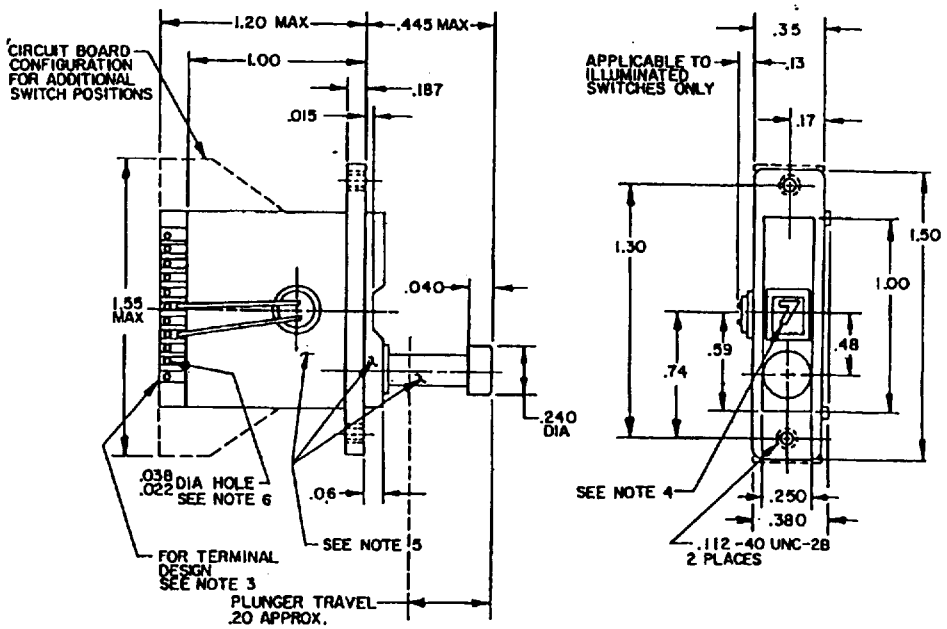
30 September 1986
SUPERSEDING
MIL-S-22710/23B
11 January 1983

PERFORMANCE SPECIFICATION SHEET

SWITCHES, CODE, INDICATING WHEEL, (PRINTED CIRCUIT), (PUSHBUTTON, UNIDIRECTIONAL, SEALED OR UNSEALED, EMI/RFI SHIELDED, ILLUMINATED (SIDE RELAMPABLE) OR NON ILLUMINATED), STYLE SRPC23

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22710.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.015	.38	.13	3.3	.350	8.89	1.00	25.4
.026	.66	.17	4.3	.380	9.65	1.20	30.5
.038	.97	.187	4.75	.445	11.30	1.30	33.0
.040	1.02	.20	5.1	.48	12.2	1.50	38.1
.06	1.5	.240	6.10	.59	15.0	1.55	39.4
.100	2.54	.250	6.35	.74	18.8		

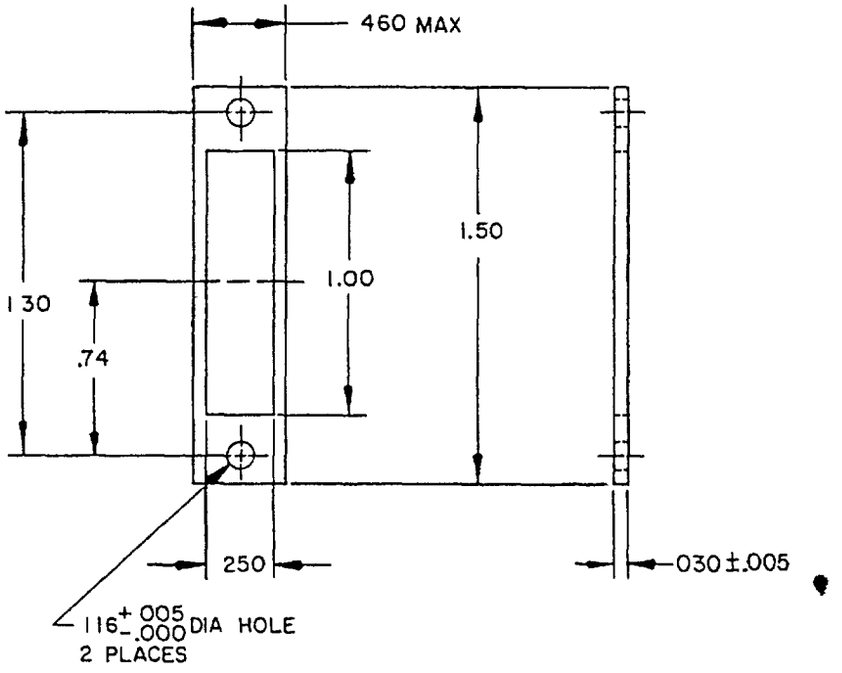
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 (.51 mm) for two place decimals and ±.010(.25 mm) for three place decimals.
4. Character height is .156(3.96 mm).
5. Housing, mounting flange and actuator are metal.
6. This size hole will accommodate one AWG wire size 22.

FIGURE 1. Switch assembly.

© denotes changes

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INCHES	MM
.005	.13
.030	.76
.116	2.95
.250	6.35
.460	11.68
.74	18 8
1.00	25.4
1 30	33.0
1 50	38.1

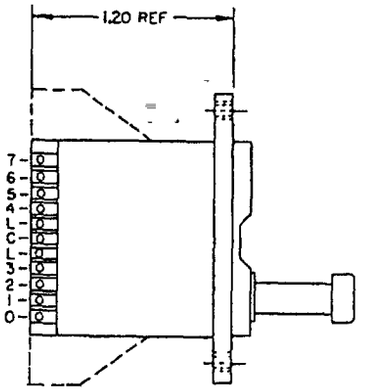
NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02(.51 mm) for two place decimals and ±.010(.25 mm) for three place decimals
4. When panel seal is specified (see table I), the material shall be silicone rubber.

FIGURE 2 Panel seal gasket.

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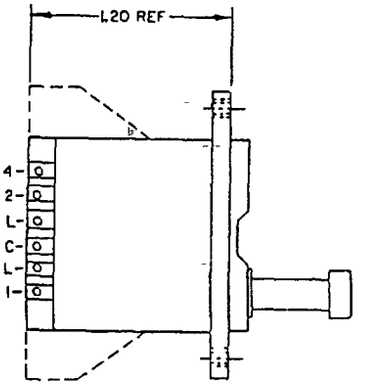
NO 1

DIAL READS	COMMON C CONNECTED TO TERMINALS							
	0	1	2	3	4	5	6	7
0	●							
1		●						
2			●					
3				●				
4					●			
5						●		
6							●	
7								●



NO 2

DIAL READS	COMMON C CONNECTED TO TERMINALS		
	1	2	4
0			
1	●		
2		●	
3	●	●	
4			●
5	●		●
6		●	●
7	●	●	●



NO 3

DIAL READS	COMMON C CONNECTED TO TERMINAL			
	0	1	2	4
0	●			
1		●		
2			●	
3		●	●	
4				●
5		●		●
6			●	●
7		●	●	●

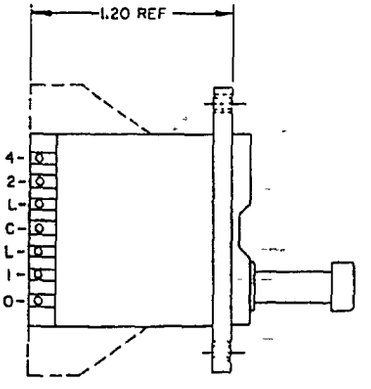
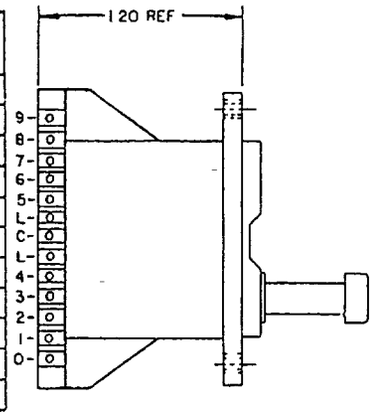


FIGURE 3 Terminal design and configuration

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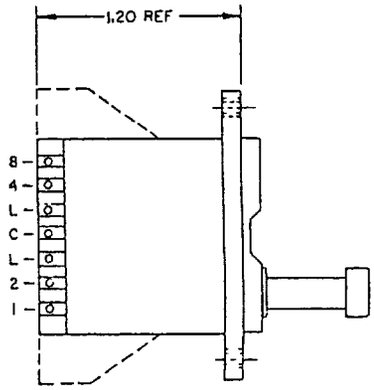
NO 4

DIAL READS	COMMON C CONNECTED TO TERMINAL									
	0	1	2	3	4	5	6	7	8	9
0	●									
1		●								
2			●							
3				●						
4					●					
5						●				
6							●			
7								●		
8									●	
9										●



NO 5

DIAL READS	COMMON C CONNECTED TO TERMINAL			
	1	2	4	8
0				
1	●			
2		●		
3	●	●		
4			●	
5	●		●	
6		●	●	
7	●	●	●	
8				●
9	●			●



NO 6

DIAL READS	COMMON C CONNECTED TO TERMINAL				
	1	2	4	8	P
0					●
1	●				
2		●			
3	●	●			●
4			●		
5	●		●		●
6		●	●		●
7	●	●	●		
8				●	
9	●			●	●

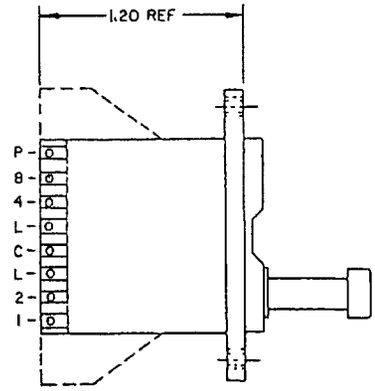
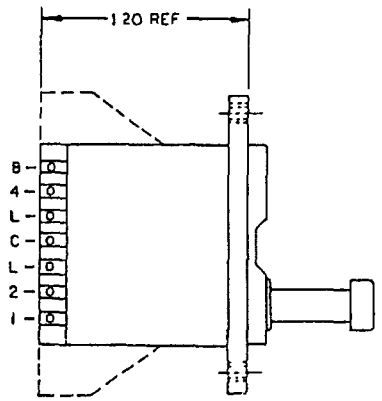


FIGURE 3 Terminal design and configuration - Continued

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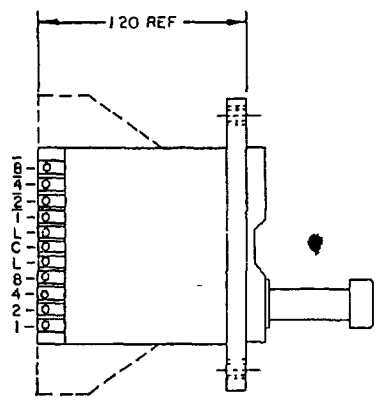
NO.7

DIAL READS	COMMON C CONNECTED TO TERMINAL			
	1	2	4	8
0	●	●		
1			●	
2	●		●	
3		●	●	
4	●	●	●	
5				●
6	●			●
7		●		●
8	●	●		●
9			●	●



NO 8

DIAL READS	COMMON C CONNECTED TO TERMINAL							
	1	2	4	8	$\bar{1}$	$\bar{2}$	$\bar{4}$	$\bar{8}$
0					●	●	●	●
1	●					●	●	●
2		●			●		●	●
3	●	●					●	●
4			●		●	●		●
5	●		●			●		●
6		●	●		●			●
7	●	●	●					●
8				●	●	●	●	
9	●			●		●	●	



NO 9

DIAL READS	COMMON C CONNECTED TO TERMINAL											
	0	1	2	3	4	5	6	7	8	9	10	11
0	●											
1		●										
2			●									
3				●								
4					●							
5						●						
6							●					
7								●				
8									●			
9										●		
10											●	
11												●

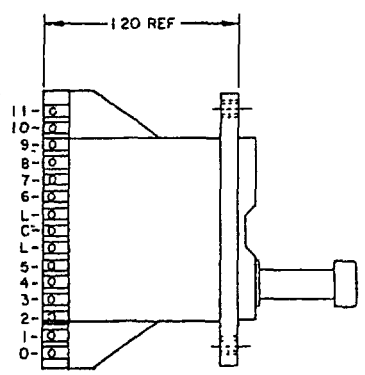
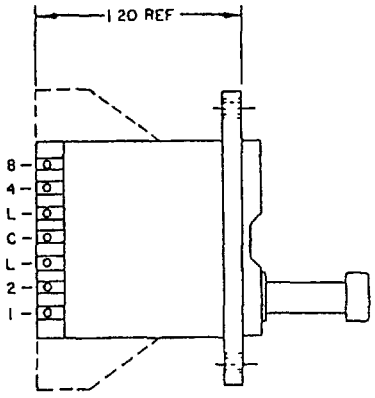


FIGURE 3 Terminal design and configuration - Continued

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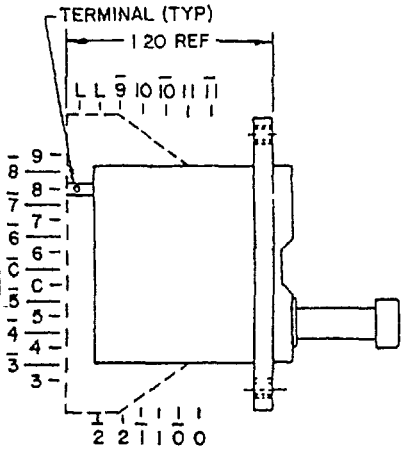
NO 10

DIAL READS	COMMON C CONNECTED TO TERMINAL			
	1	2	4	8
0				
1	●			
2		●		
3	●	●		
4			●	
5	●		●	
6		●	●	
7	●	●	●	
8				●
9	●			●
10		●		●
11	●	●		●



NO 11 2 POLE

DIAL READS	COMMON C CONNECTED TO TERMINAL										DIAL READS	COMMON C CONNECTED TO TERMINAL									
	C	1	2	3	4	5	6	7	8	9		C	1	2	3	4	5	6	7	8	9
C	●										C	●									
1		●									1		●								
2			●								2			●							
3				●							3				●						
4					●						4					●					
5						●					5						●				
6							●				6							●			
7								●			7								●		
8									●		8									●	
9										●	9										●
10											10										
11											11										



NO 12 2 POLE

DIAL READS	COMMON C CONNECTED TO TERMINAL									DIAL READS	COMMON C CONNECTED TO TERMINAL								
	C	1	2	3	4	5	6	7	8		C	1	2	3	4	5	6	7	8
C	●									C	●								
1		●								1		●							
2			●							2			●						
3				●						3				●					
4					●					4					●				
5						●				5						●			
6							●			6							●		
7								●		7								●	
8									●	8									●
9										9									

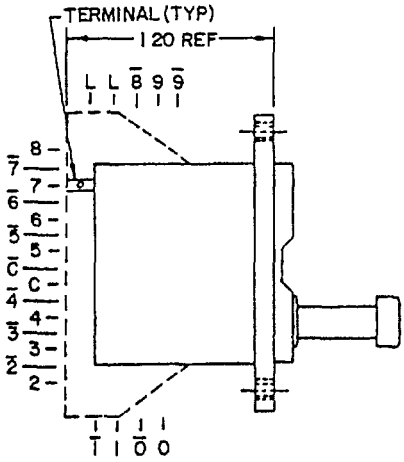


FIGURE 3 Terminal design and configuration - Continued

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NO 13 2 POLE

DIAL READS	COMMON C CONNECTED TO TERMINAL								DIAL READS	COMMON $\overline{C}$ CONNECTED TO TERMINAL							
	0	1	2	3	4	5	6	7		$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$	$\overline{4}$	$\overline{5}$	$\overline{6}$	$\overline{7}$
0	●								0	●							
1		●							1		●						
2			●						2			●					
3				●					3				●				
4					●				4					●			
5						●			5						●		
6							●		6							●	
7								●	7								●

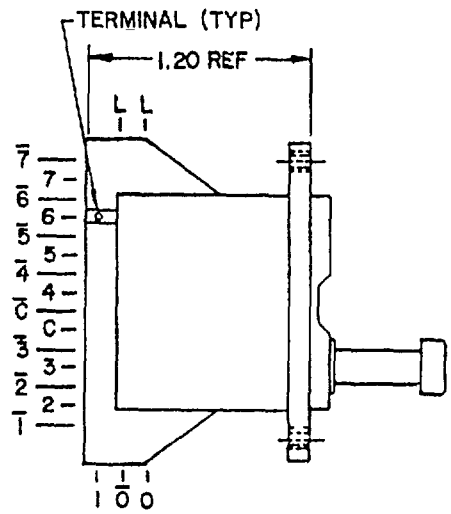


FIGURE 3. Terminal design and configuration - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Construction. D

Angle of throw: 30°, 36°, or 45° indexing, also octal or binary codes.

Temperature-life characteristic: D, test loads as shown in table I.

TABLE I. Electrical test load.

Test or condition	Resistive	
At atmospheric pressure	0.125 A 0.125 A	28 V dc 115 V ac
At reduced barometric pressure	0.05 A	28 V dc and 115 V ac
Dry circuit	10 mA	50 mV

Vibration grade: 3 (15g, 10-2, 000 Hz).

Shock types: M (100 g's).

Altitude rating: C (70,000 feet).

Terminal design: See figure 3 and table III.

Contact resistance:

Initial: 50 milliohms, maximum.

Following life (rotational): 70 milliohms, maximum.

Luminance: Illuminated switches only as shown in table III.

TABLE II. Luminance.

Lamp	Switch filter	Brightness	Rating	
			Volts	Ampere
MS90452-6833	Clear	1.50 ±.75 foot-lamberts	5	.060
MS90452-6833	Red	.75 ±.50 foot-lamberts	5	.060

Rotational torque: 20 ounce minimum to 40 ounce maximum.

Seal: Method II, when specified in table I.

Dielectric withstanding voltage: 1,000 volts, rms, at atmospheric pressure and 450 volts, rms, at reduced barometric pressure applied between the switch terminals and terminals to case.

Sand and dust: Applicable.

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Explosion: Applicable.

Life (rotational): Test condition G, method 206 of MIL-STD-202.

© EMI/RFI shielding: Applicable when specified (see table III).

Military part number: M22710/23- (dash number from table III).

Supersession data: This style supersedes style SRPC09 of MIL-C-22710/9.

© TABLE III. Part numbers and characteristics.

Dash number M22710/23-		Terminal design and circuit configuration number	Seal test	Illumination	Panel seal (see figure 2)	Readout characters
With EMI/RFI shielding	Without EMI/RFI shielding					
1001	001	1 of figure 3	No	None	No	B/W
1002	002		No	None	No	W/B
1003	003		Yes	None	No	B/W
1004	004		Yes	None	No	W/B
1005	005		Yes	None	Yes	B/W
1006	006		Yes	None	Yes	W/B
1007	007		No	White	No	B/W
1008	008		Yes	White	No	B/W
1009	009		Yes	White	Yes	B/W
1010	010		No	Red	No	B/W
1011	011		Yes	Red	No	B/W
1012	012		Yes	Red	Yes	B/W
1013	013	2 of figure 3	No	None	No	B/W
1014	014		No	None	No	W/B
1015	015		Yes	None	No	B/W
1016	016		Yes	None	No	W/B
1017	017		Yes	None	Yes	B/W
1018	018		Yes	None	Yes	W/B
1019	019		No	White	No	B/W
1020	020		Yes	White	No	B/W
1021	021		Yes	White	Yes	B/W
1022	022		No	Red	No	B/W
1023	023		Yes	Red	No	B/W
1024	024		Yes	Red	Yes	B/W
1025	025	3 of figure 3	No	None	No	B/W
1026	026		No	None	No	W/B
1027	027		Yes	None	No	B/W
1028	028		Yes	None	No	W/B
1029	029		Yes	None	Yes	B/W
1030	030		Yes	None	Yes	W/B
1031	031		No	White	No	B/W
1032	032		Yes	White	No	B/W
1033	033		Yes	White	Yes	B/W
1034	034		No	Red	No	B/W
1035	035		Yes	Red	No	B/W
1036	036		Yes	Red	Yes	B/W

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© TABLE III. Part numbers and characteristics - Continued.

Dash number M22710/23-		Terminal design and circuit configuration number	Seal test	Illumination	Panel seal (see figure 2)	Readout characters
With EMI/RFI shielding	Without EMI/RFI shielding					
1037	037	4 of figure 3	No	None	No	B/W
1038	038		No	None	No	W/B
1039	039		Yes	None	No	B/W
1040	040		Yes	None	No	W/B
1041	041		Yes	None	Yes	B/W
1042	042		Yes	None	Yes	W/B
1043	043		No	White	No	B/W
1044	044		Yes	White	No	B/W
1045	045		Yes	White	Yes	B/W
1046	046		No	Red	No	B/W
1047	047		Yes	Red	No	B/W
1048	048		Yes	Red	Yes	B/W
1049	049	5 of figure 3	No	None	No	B/W
1050	050		No	None	No	W/B
1051	051		Yes	None	No	B/W
1052	052		Yes	None	No	W/B
1053	053		Yes	None	Yes	B/W
1054	054		Yes	None	Yes	W/B
1055	055		No	White	No	B/W
1056	056		Yes	White	No	B/W
1057	057		Yes	White	Yes	B/W
1058	058		No	Red	No	B/W
1059	059		Yes	Red	No	B/W
1060	060		Yes	Red	Yes	B/W
1061	061	6 of figure 3	No	None	No	B/W
1062	062		No	None	No	W/B
1063	063		Yes	None	No	B/W
1064	064		Yes	None	No	W/B
1065	065		Yes	None	Yes	B/W
1066	066		Yes	None	Yes	W/B
1067	067		No	White	No	B/W
1068	068		Yes	White	No	B/W
1069	069		Yes	White	Yes	B/W
1070	070		No	Red	No	B/W
1071	071		Yes	Red	No	B/W
1072	072		Yes	Red	Yes	B/W
1073	073	7 of figure 3	No	None	No	B/W
1074	074		No	None	No	W/B
1075	075		Yes	None	No	B/W
1076	076		Yes	None	No	W/B
1077	077		Yes	None	Yes	B/W
1078	078		Yes	None	Yes	W/B
1079	070		No	White	No	B/W
1080	080		Yes	White	No	B/W
1081	081		Yes	White	Yes	B/W
1082	082		No	Red	No	B/W
1083	083		Yes	Red	No	B/W
1084	084		Yes	Red	Yes	B/W

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© TABLE III. Part numbers and characteristics - Continued.

Dash number M22710/23-		Terminal design and circuit configuration number	Seal test	Illumination	Panel seal (see figure 2)	Readout characters
With EMI/RFI shielding	Without EMI/RFI shielding					
1085	085		No	None	No	B/W
1086	086		No	None	No	W/B
1087	087		Yes	None	No	B/W
1088	088		Yes	None	No	W/B
1089	089		Yes	None	Yes	B/W
1090	090	8 of figure 3	Yes	None	Yes	W/B
1091	091		No	White	No	B/W
1092	092		Yes	White	No	B/W
1093	093		Yes	White	Yes	B/W
1094	094		No	Red	No	B/W
1095	095		Yes	Red	No	B/W
1096	096		Yes	Red	Yes	B/W
1097	097		No	None	No	B/W
1098	098		No	None	No	W/B
1099	099		Yes	None	No	B/W
1100	100		Yes	None	No	W/B
1101	101		Yes	None	Yes	B/W
1102	102	9 of figure 3	Yes	None	Yes	W/B
1103	103		No	White	No	B/W
1104	104		Yes	White	No	B/W
1105	105		Yes	White	Yes	B/W
1106	106		No	Red	No	B/W
1107	107		Yes	Red	No	B/W
1108	108		Yes	Red	Yes	B/W
1109	109		No	None	No	B/W
1110	110		No	None	No	W/B
1111	111		Yes	None	No	B/W
1112	112		Yes	None	No	W/B
1113	113		Yes	None	Yes	B/W
1114	114	10 of figure 3	Yes	None	Yes	W/B
1115	115		No	White	No	B/W
1116	116		Yes	White	No	B/W
1117	117		Yes	White	Yes	B/W
1118	118		No	Red	No	B/W
1119	119		Yes	Red	No	B/W
1120	120		Yes	Red	Yes	B/W
1121	121		No	None	No	B/W
1122	122		No	None	No	W/B
1123	123		Yes	None	No	B/W
1124	124		Yes	None	No	W/B
1125	125		Yes	None	Yes	B/W
1126	126	11 of figure 3	Yes	None	Yes	W/B
1127	127		No	White	No	B/W
1128	128		Yes	White	No	B/W
1129	129		Yes	White	Yes	B/W
1130	130		No	Red	No	B/W
1131	131		Yes	Red	No	B/W
1132	132		Yes	Red	Yes	B/W

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© TABLE III. Part numbers and characteristics - Continued.

Dash number M22710/23-		Terminal design and circuit configuration number	Seal test	Illumination	Panel seal (see figure 2)	Readout characters
With EMI/RFI shielding	Without EMI/RFI shielding					
1133	133		No	None	No	B/W
1134	134		No	None	No	W/B
1135	135		Yes	None	No	B/W
1136	136		Yes	None	No	W/B
1137	137		Yes	None	Yes	B/W
1138	138	12 of figure 3	Yes	None	Yes	W/B
1139	139		No	White	No	B/W
1140	140		Yes	White	No	B/W
1141	141		Yes	White	Yes	B/W
1142	142		No	Red	No	B/W
1143	143		Yes	Red	No	B/W
1144	144		Yes	Red	Yes	B/W
1145	145		No	None	No	B/W
1146	146		No	None	No	W/B
1147	147		Yes	None	No	B/W
1148	148		Yes	None	No	W/B
1149	149		Yes	None	Yes	B/W
1150	150	13 of figure 3	Yes	None	Yes	W/B
1151	151		No	White	No	B/W
1152	152		Yes	White	No	B/W
1153	153		Yes	White	Yes	B/W
1154	154		No	Red	No	B/W
1155	155		Yes	Red	No	B/W
1156	156		Yes	Red	Yes	B/W

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - SM
Navy - AS
Air Force - 99
DLA - ES

Review activities:
Army - AR, ME
Navy - MC
Air Force - 11, 17

Preparing activity:
Navy - EC

Agent:
DLA - Es

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